

Portfolio Project Documentation

Developer: Muhammad Atif

Creative Web Developer | Video Editor | Frontend Developer

1. PROJECT OVERVIEW

This project is a modern, interactive 3D portfolio website built for Muhammad Atif. It showcases web development projects, video editing experience, skillset, and career background. The application utilizes high-performance 3D graphics (Three.js/WebGL), smooth GSAP scroll animations, and a customized Codentra green & dark steel theme design.

2. COMPLETE TECHNOLOGIES USED

Frontend Core	React v18.3.1 & TypeScript v5.5	UI component architecture, state management, and type safetv.
Build Tool	Vite v5.4.1	Fast HMR development server and optimized bundle compilation.
3D Graphics	Three.js v0.168.0 & WebGL	Core 3D WebGL rendering engine for interactive models.
React 3D Library	@react-three/fiber & @react-three/drei	React component wrappers for 3D lights, cameras, HDR environment. and GLTF models.
Post Processing	@react-three/postprocessing	Advanced 3D shader post-processing visual effects.
Animation Engine	GSAP v3.12.7 & ScrollTrigger	Smooth scroll animations, timelines, and text splitting effects.
3D Compression	Draco Decoder	Decodes compressed 3D GLTF files for rapid initial page load.
UI Utilities	react-icons & react-fast-marquee	Social icons, navigation arrows, and continuous marquee banner.
Analytics	@vercel/analytics	Real-time website traffic metrics tracking.
Styling & Design	HTML5, CSS Variables, Flexbox/Grid	Custom Codentra theme with dark background and glowing green accents.

3. KEY FEATURES & FUNCTIONALITY

- Interactive 3D Model Scene: Rendered using React Three Fiber with custom lighting and HDR environment background.
- Smooth Scroll Animations: GSAP ScrollTrigger drives element reveal animations across Landing, Work, and About sections.
- Codentra Brand Theme: Customized logo branding with transparent SVG, dark steel gray tone, and glowing green accents.
- Video Editing & Web Experience Showcase: Highlighted career roles at My Impact Meter and Encho Software.
- Fully Responsive Layout: Dynamic scaling across Desktop, Laptop, Tablet, and Mobile viewports.
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